Duality in String Theory

Following Witten, let's start with one of the five string theories, say the Type I string, and imagine that all of its nine space dimensions are flat and unfurled. This, of course, is not at all realistic, but it makes the discussion simpler; we will return to curled-up dimensions shortly. We begin by assuming that the string coupling constant is much less than 1. In this case, perturbative tools are valid, and hence many of the detailed properties of the theory can and have been worked out with accuracy. If we increase the value of the coupling constant but still keep it a good deal less than 1, perturbative methods can still be used. The detailed properties of the theory will change somewhat—for instance, the numerical values associated with the scattering of one string off another will be a bit different because the multiple loop processes of Figure 12.6 make greater contributions when the coupling constant increases. But beyond these changes in detailed numerical properties, the overall physical content of the theory remains the same, so long as the coupling constant stays in the perturbative realm.

As we increase the Type I string coupling constant beyond the value 1, perturbative methods become invalid and so we focus only on the limited set of nonperturbative masses and charges—the BPS states—that are still within our ability to understand. Here is what Witten argued, and later confirmed through joint work with Joe Polchinski of the University of California at Santa Barbara: These strong coupling characteristics of Type I string theory exactly agree with known properties of Heterotic-O string theory, when the latter has a small value for its string coupling constant. That is, when the coupling constant of the Type I string is large, the particular masses and charges that we know how to extract are precisely equal to those of the Heterotic-O string when its coupling constant is small. This gives us a strong indication that these two string theories, which at first sight, like water and ice, seem totally different, are actually dual. It persuasively suggests that the physics of the Type I theory for large values of its coupling constant is identical to the physics of the Heterotic-O theory for small values of its coupling constant. Related arguments gave equally persuasive evidence that the reverse is also true: The physics of the Type I theory for small values of its coupling constant is identical to that of the Heterotic-O theory for large values of its